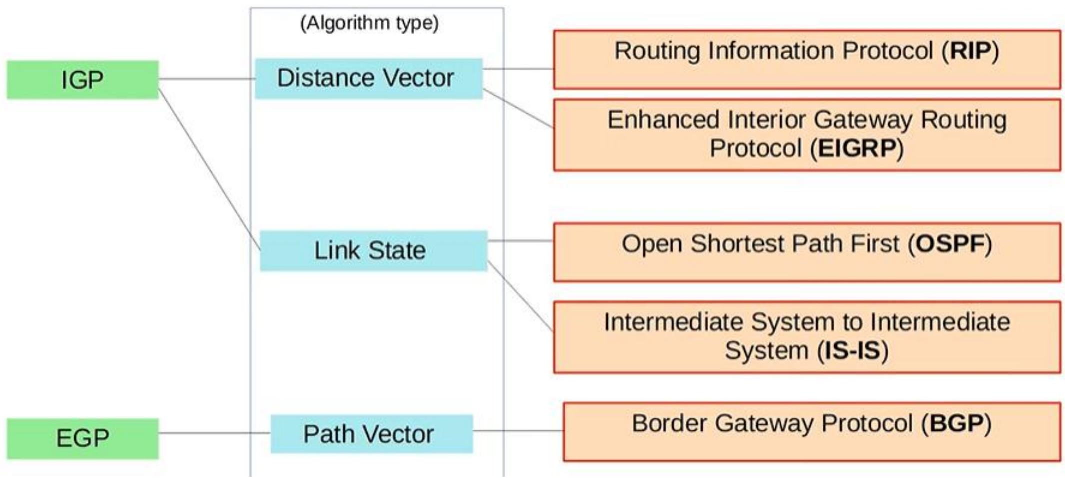
OSPF (Part-1)

- Topics covered:
    - Basic OSPF Operations (introduction)
    - OSPF Areas
    - Basic OSPF Configuration

OSPF – Introduction, LSAs, LSDB, and Areas

Previous Overview

Types of Dynamic Routing Protocols:



Link State Routing Protocols:

- When using a **link state** routing protocol, every router creates a 'connectivity map' of the network.
- To allow this, each router advertises information about its interfaces (connected networks) to its neighbors. These advertisements are passed along to other routers, until all routers in the network develop the same map of the network.
- Each router independently uses this map to calculate the best routes to each destination.
- Link state protocols use more resources (CPU) on the router, because more information is shared.
- However, link state protocols tend to be faster in reacting to changes in the network than distance vector protocols.

What is OSPF?

OSPF stands for Open Shortest Path First.

The OSPF protocol uses the Shortest Path First algorithm, created by Dutch computer scientist **Edsger Dijkstra**.

Another name for the algorithm is **Dijkstra's algorithm**.
Remember that name — it could be an exam question.

OSPF Versions:

There are three versions of OSPF.

- Version 1
  Released in 1989.
  It is old and not in use anymore.

- Version 2
  Released in 1998.
  This is typically used in IPv4 networks.

  The exam topics list OSPF version 2, so when OSPF is discussed here, it refers to OSPFv2.

- Version 3
  Developed for IPv6.
  It can also be used for IPv4, but version 2 is more common in IPv4 networks.

General OSPF Concepts: LSA and LSDB

Routers store information about the network in **LSAs (Link State Advertisements)**.

These LSAs are organized in a structure called the **LSDB (Link State Database)**.

LSA and LSDB are two important OSPF terms and will appear throughout OSPF discussions.

Routers flood LSAs until all routers in the same OSPF area develop the same network map, meaning the same LSDB.

Important terms introduced here:

- LSA
- LSDB
- Flood

- Area

Flooding in OSPF means sending LSAs to all OSPF neighbours, like how switches flood broadcasts or unknown unicasts.

OSPF areas are a unique aspect of OSPF and will be explained later in the video.

LSA and LSDB Example (4-Router Network):

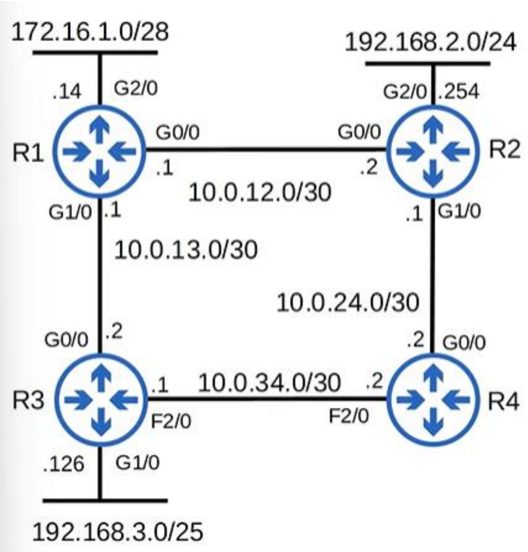
Let's elaborate on LSAs and the LSDB using an example.

Assume this network of four routers is running OSPF.

All routers:

- Are OSPF neighbours
- Have the same LSDB
- The network is stable

Network Diagram (Before Change):

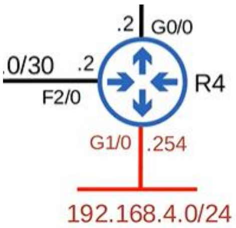


Enabling OSPF on R4 G1/0:

OSPF is enabled on R4's G1/0 interface.

R4 must inform other routers about this new network segment.

So, R4 creates an LSA describing the network on G1/0.



Contents of the LSA:

An OSPF LSA contains more information, but basic fields include:

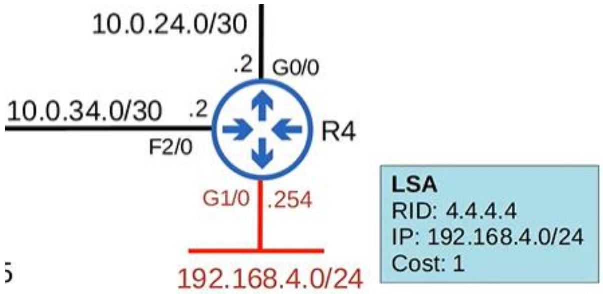
- **Router ID (RID)**
  For this demonstration, R4's RID is 4.4.4.4.

  None of R4's physical interfaces use this IP address, so:

  - Either a loopback interface exists with IP 4.4.4.4
  - Or the router ID was manually configured

- **Network information**
  The network on G1/0 is included because that is the purpose of the LSA.

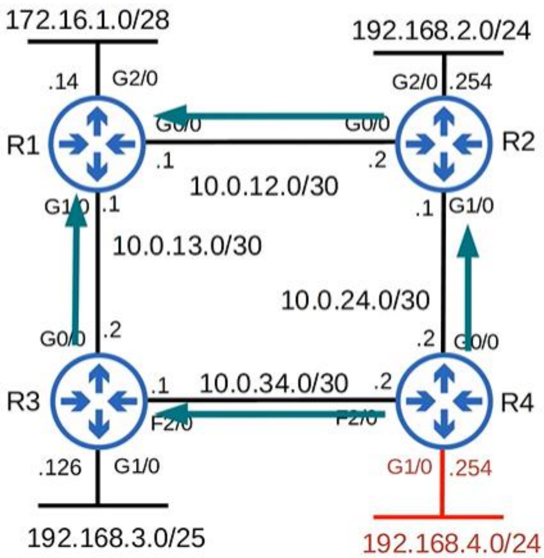
- **Cost**
  R4's cost is included.
  The Gigabit Ethernet interface has a cost of 1.



LSA Flooding Process:

The LSA is flooded throughout the network until all routers receive a copy.

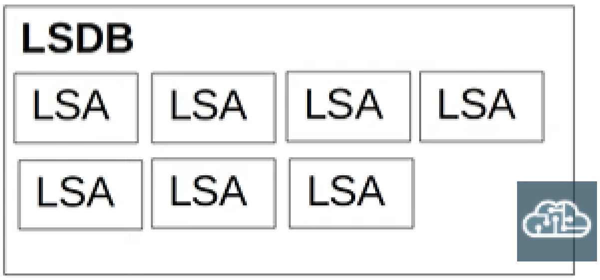
Flooding Visualization



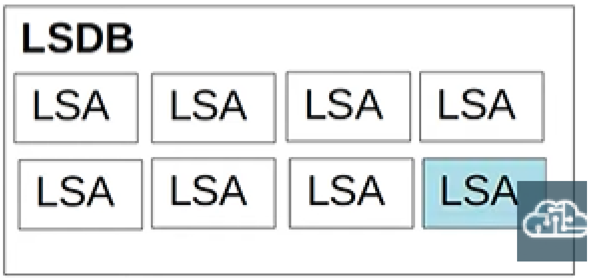
LSDB After Flooding:

As a result, all routers in the OSPF area have the same LSDB.

LSDB Conceptual View Before G1/0 interface active



Now that OSPF is activated on R4 G1/0, that LSA is added to the LSDB.



This LSDB is identical on all routers in the same OSPF area.

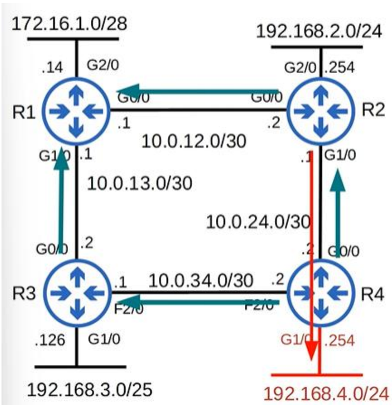
Route Calculation Using SPF:

Each router now runs the SPF (Dijkstra's) algorithm to calculate the best route to 192.168.4.0/24.

Each router has a complete map of the network.

For example:

- We can see that R2's best route to 192.168.4.0/24 is via G1/0
- R2 sees the same topology logically and calculates the same result



Routers do not see diagrams visually, but functionally it is equivalent.

LSA Aging Timer:

Each individual LSA has an aging timer.

- Default aging timer: 30 minutes
- When the timer expires, the LSA is flooded again
- This happens once every 30 minutes by default

Summary: OSPF LSA Process

There are three main steps in OSPF for sharing LSAs and determining best routes.

Step 1: Neighbour Formation

Routers become neighbours with other routers on the same segment.

Example:

- R4 becomes neighbours with R2 and R3

Step 2: LSA Exchange

Routers exchange LSAs with their neighbours.

Step 3: SPF Calculation

Each router:

- Independently calculates best routes
- Inserts them into the routing table

These steps will be covered in depth later.

OSPF Areas

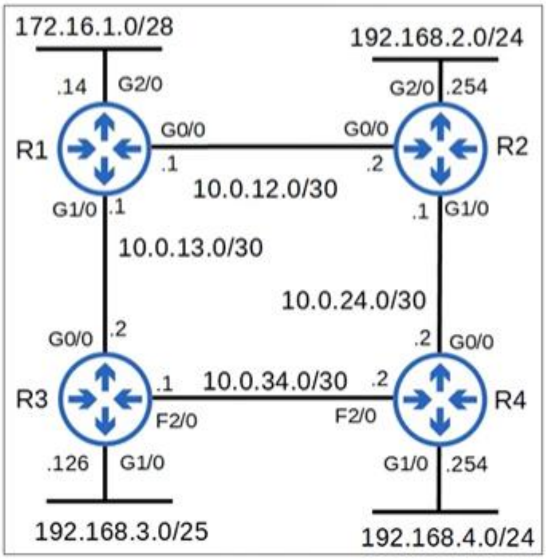
Introduction to OSPF Areas

OSPF areas are a fundamental concept not found in RIP or EIGRP.

OSPF uses areas to divide up the network.

Small networks can be single-area without performance issues.

Small Network Example (Single Area):



(All routers in one area)

No performance degradation occurs in this case.

Why Multiple Areas Are Needed:

In large networks, a single-area design can cause problems.

Example:

- 500 routers
- Over 1000 subnets

Negative effects:

- SPF calculations take longer
- Exponentially more CPU usage
- Large LSDB consumes more memory

- Any small change floods LSAs to all routers
- All routers must recalculate SPF

Dividing the network into **smaller areas** avoids these problems.

Exam Scope Note:

The exam topics mention only single-area OSPF.

Therefore:

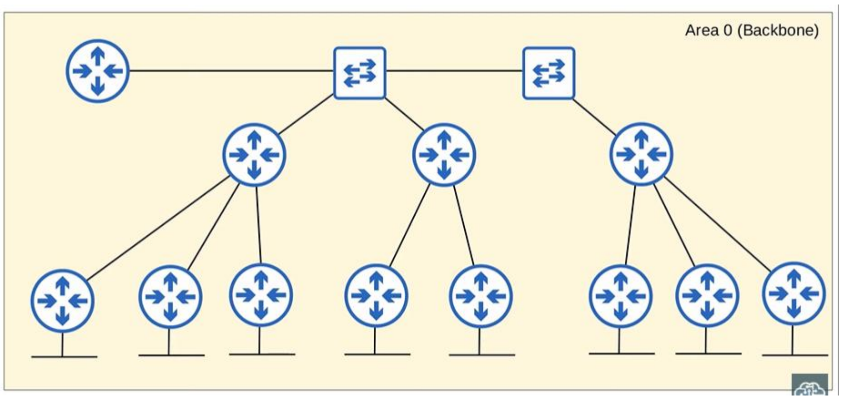
- Only a brief overview of OSPF areas is required
- Deep multi-area design details are not required for CCNA

Backbone Area (Area 0)

A large network can be designed as a single area.

All interfaces are placed in Area 0, also called the Backbone Area.

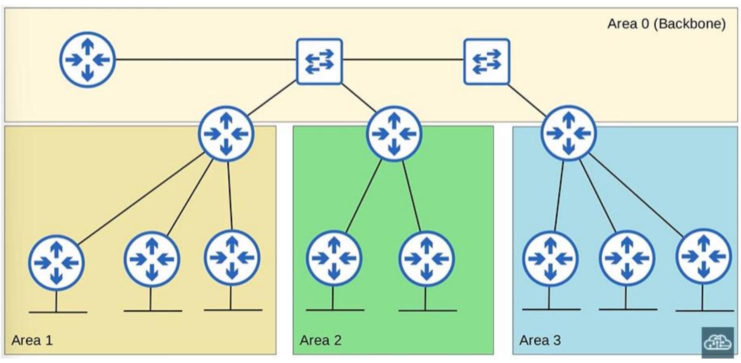
Single-Area Backbone Example



Area 0 is of special importance in OSPF.

Dividing the Network into Multiple Areas:

Instead of one large area, the network can be divided.



What is an Area?

An area is:

- A set of routers and links
- That share the same LSDB

In the diagram:

- Area 0
- Area 1
- Area 2
- Area 3

Total: 4 areas

Each area maintains a unique LSDB.

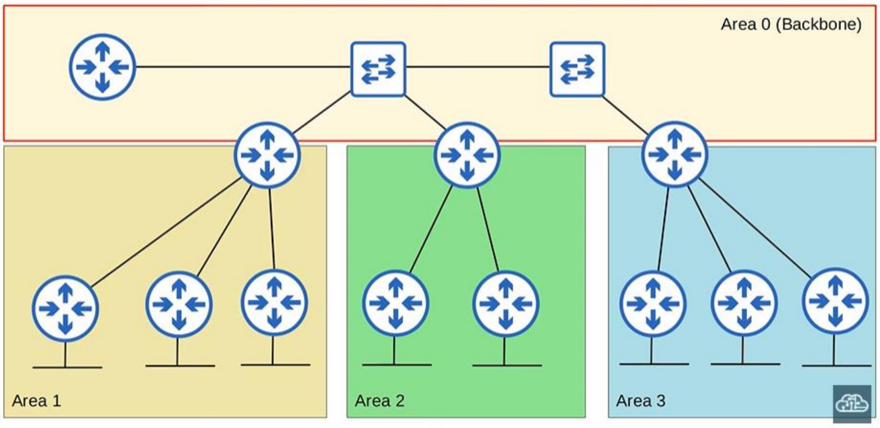
RULES

Backbone Area Rule:

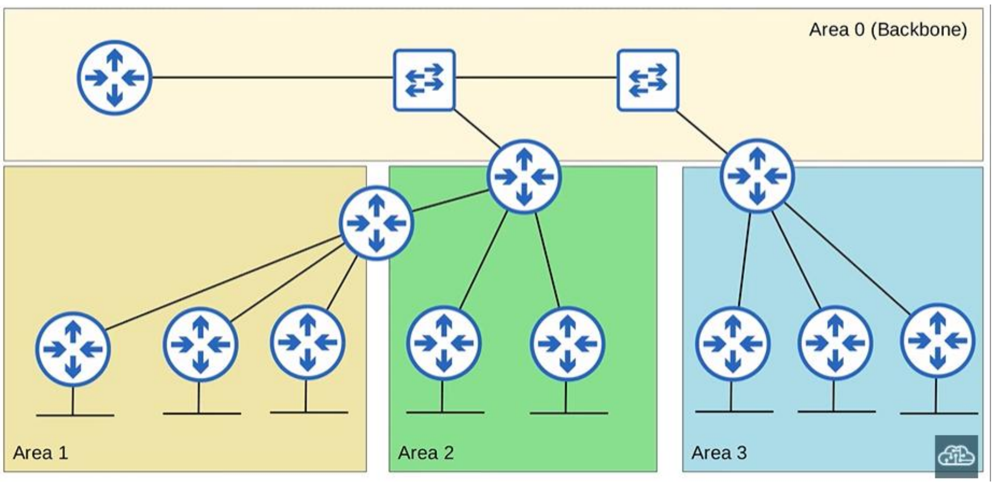
The backbone area (Area 0) is a special area.

All other areas must connect to Area 0.

Allowed Design:



Not Allowed Design:



(no direct connection between Area 1 and Area 0)

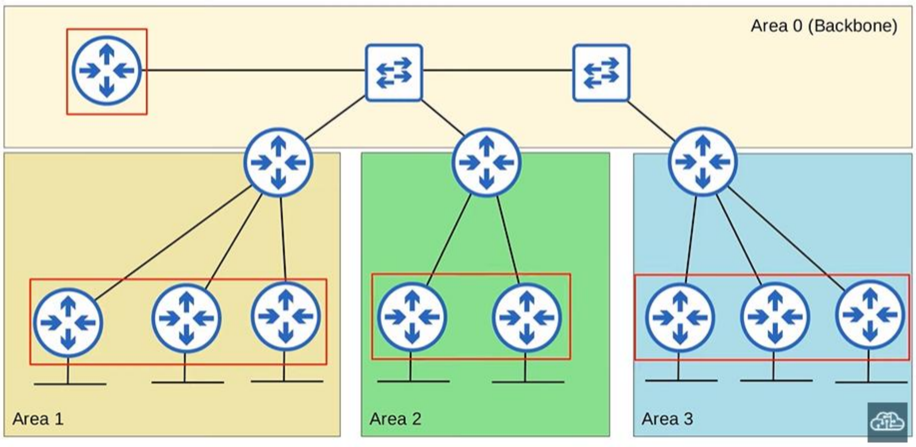
This design is not allowed in OSPF.

Internal Routers:

Routers with all interfaces in the same area are called internal routers.

Examples:

- Routers entirely in Area 0 → internal to Area 0
- Routers entirely in Area 1 → internal to Area 1
- Same for Areas 2 and 3



Internal routers do not connect multiple areas.

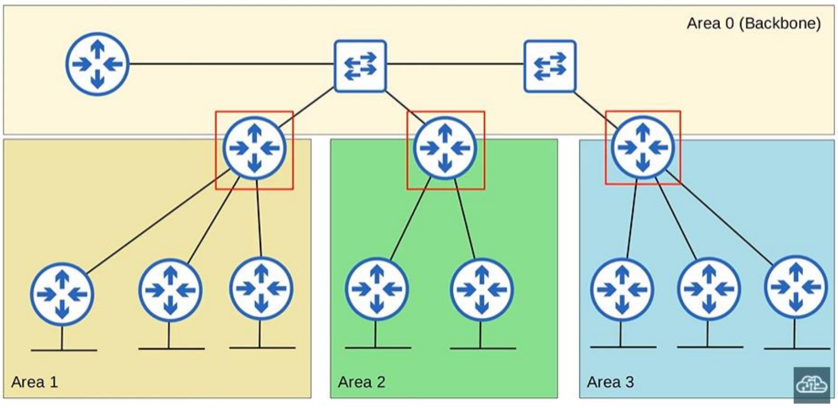
Area Border Routers (ABRs):

Routers with interfaces in multiple areas are called area border routers, because they are the border between different OSPF areas.

In this network, which routers are ABRs?

- This router, connected to area 0 and area 1, is an ABR.
- This router, connected to area 0 and area 2, is also an ABR.
- This router, connected to area 0 and area 3, is an ABR.

ABR Identification Diagram



(Routers touching Area 0 + another area are ABRs)

So remember that ABRs (Area Border Routers) are routers with interfaces in multiple OSPF areas.

ABR LSDB Behaviour and Design Recommendation:

One more bit of information about ABRs.

ABRs maintain a separate LSDB for each area they are connected to.

It is recommended that you connect an ABR to a maximum of 2 areas.

Connecting an ABR to 3 or more areas can overburden the router.

So, a design like this is good OSPF network design, with each ABR connected to only 2 areas.

Backbone Routers:

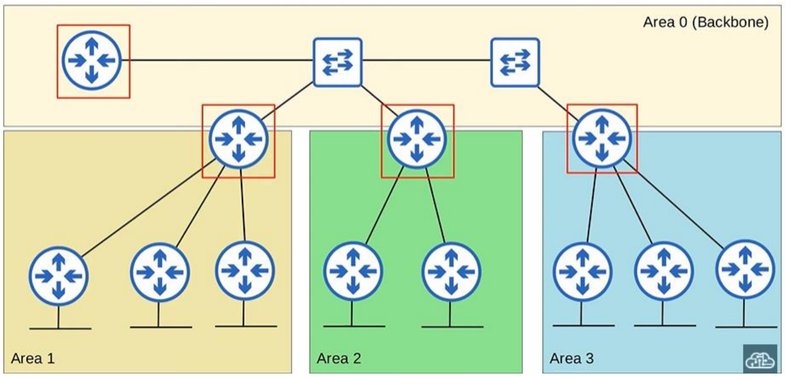
Routers connected to the backbone area, which is area 0, are called backbone routers.

This includes area border routers, by the way.

So, which routers in this network are backbone routers?

- This router is connected only to area 0, so it is a backbone router.
    - It is also an internal router, internal to area 0.
- This router is a backbone router and an ABR.
- Same for this router, and this router.
    - They are both backbone routers and area border routers (ABRs).

Backbone Router Diagram

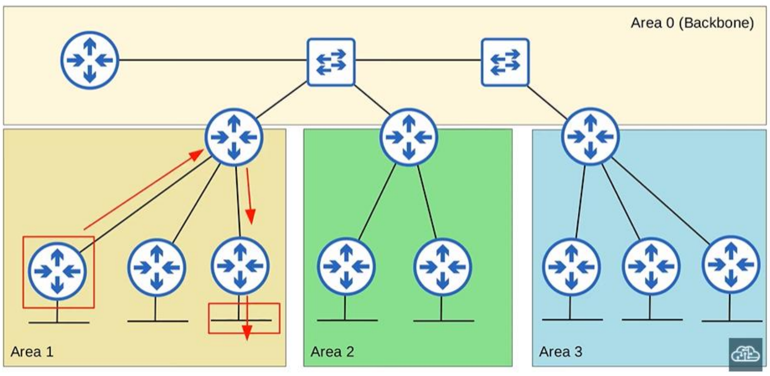


Intra-Area Routes:

An intra-area route is a route to a destination inside the same OSPF area.

For example, from a router in area 1 to a destination that is also in area 1.

Intra-Area Route Example



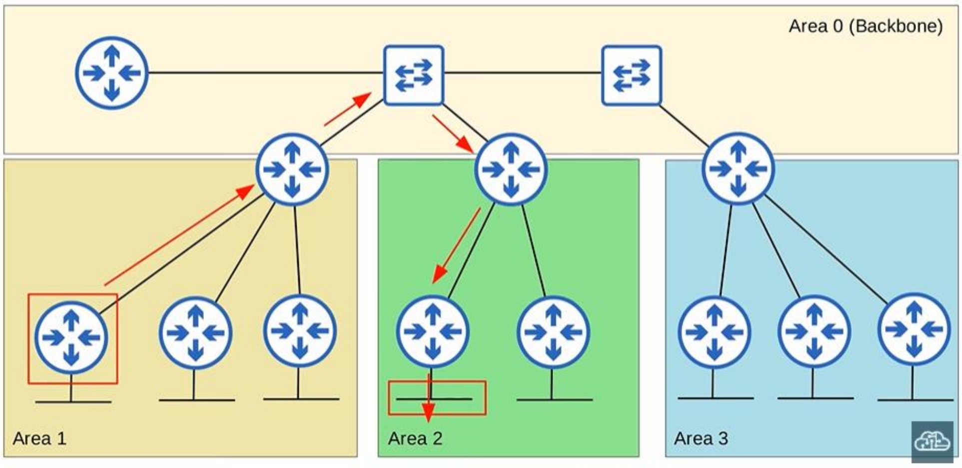
If this router learns a route to this subnet, it is considered an intra-area route, because the destination is in the same area as the router.

Interarea Routes:

An interarea route is a route to a destination in a different OSPF area.

For example, if a router in area 1 learns a route to a destination in area 2, that is an interarea route.

Interarea Route Example



If this router in area 1 learns a route to this subnet in area 2, it is considered an interarea route.

The router and the destination are in different OSPF areas.

Key OSPF Area Terms (Reminder)

So, those are some important OSPF terms regarding OSPF areas.

Make sure you learn and understand these terms:

- Area
- Backbone area
- Internal router
- Area border router
- Backbone router
- Intra-area route
- Interarea route

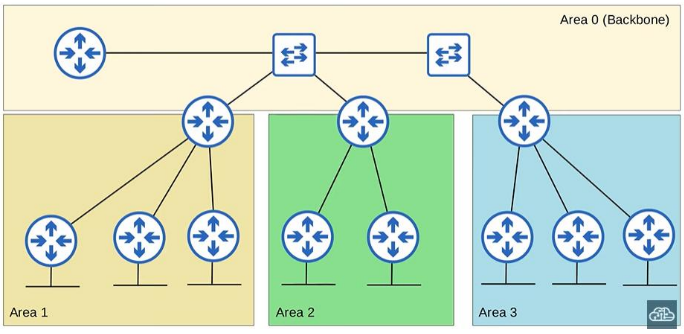
OSPF Area Rule 1: Areas Must Be Contiguous

OSPF areas should be contiguous.

What does that mean?

It means that each individual area should be connected, not divided up.

Contiguous Area Example (Correct):



All areas here are contiguous.

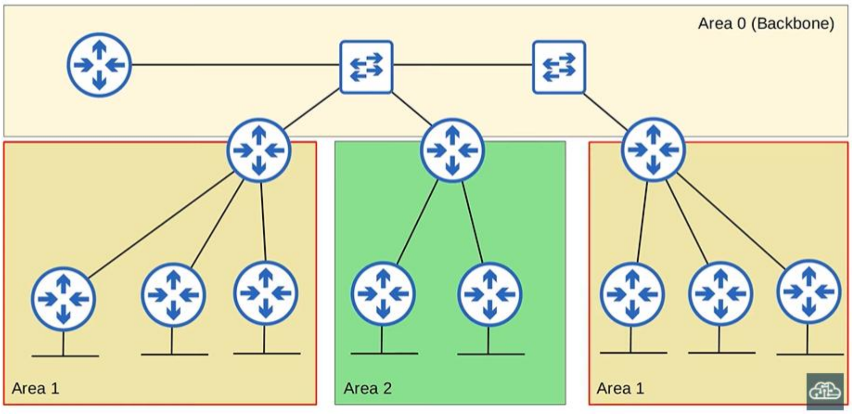
Non-Contiguous Areas (Not Allowed):

Now let's look at what it means to be non-contiguous.

Area 1 is now non-contiguous.

Half of area 1 is here, and half of area 1 is here.

Non-Contiguous Area Example (Incorrect):



This kind of network design is not allowed in OSPF and will cause problems.

Fixing Non-Contiguous Areas:

Instead of having area 1 split up like this, you should make the section on the right a separate area, area 3.

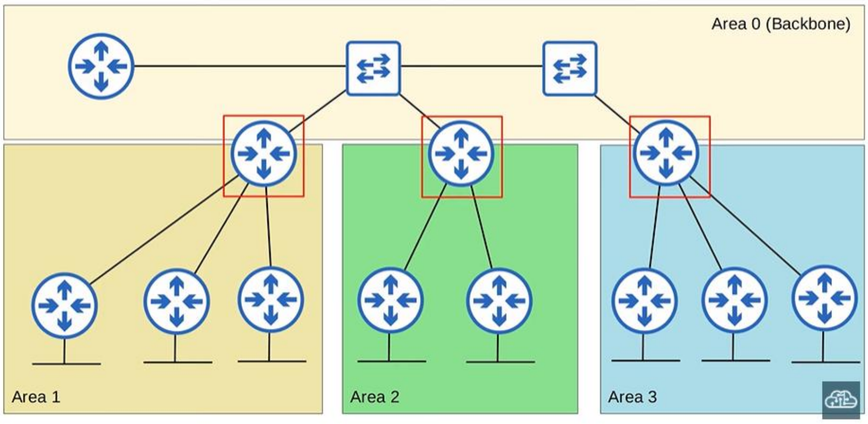
Now all areas are contiguous and OSPF can function properly.

OSPF Area Rule 2: Every Area Must Connect to Area 0

All OSPF areas must have at least one ABR connected to the backbone area.

This rule is worth repeating.

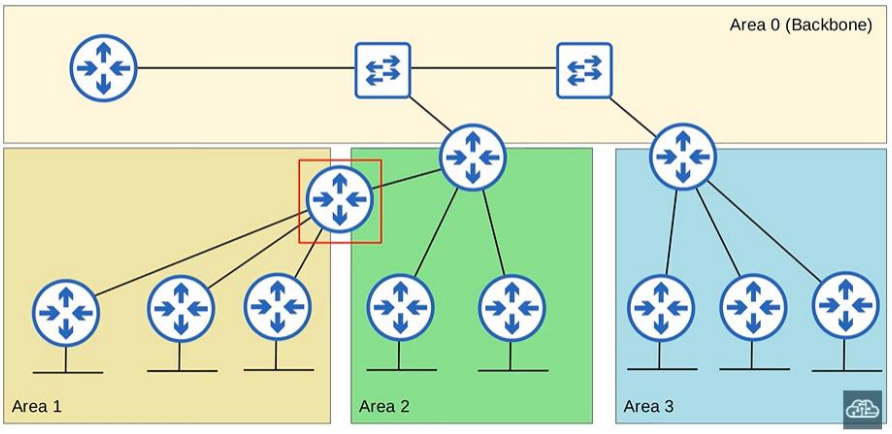
Correct Design:



Area 1, Area 2, and Area 3 all have ABRs connected to Area 0.

This is correct OSPF network design.

Incorrect Design (Not Allowed):



This is not correct OSPF design and will cause problems, because area 1 does not have an ABR connected to area 0.

OSPF Area Rule 3: Same Subnet, Same Area

OSPF interfaces in the same subnet must be in the same area.

If they are not in the same area:

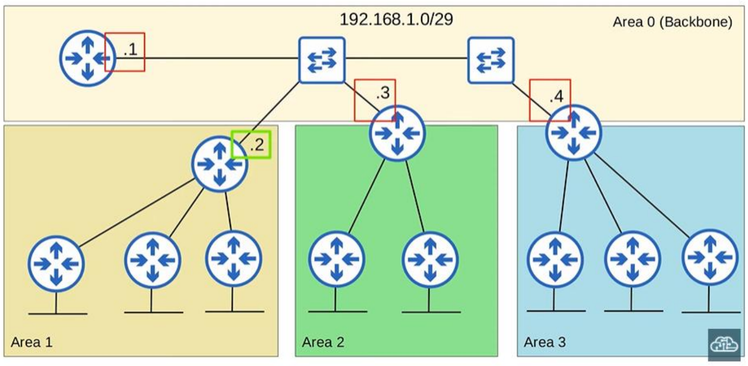
- They will not become OSPF neighbours
- They will not exchange routing information

Neighbour Formation Failure Example:

In this example, three routers have interfaces in area 0 on the 192.168.1.0/29 subnet.

This router also has an interface in the same subnet, but it is configured in area 1, not area 0.

Misconfigured Example:

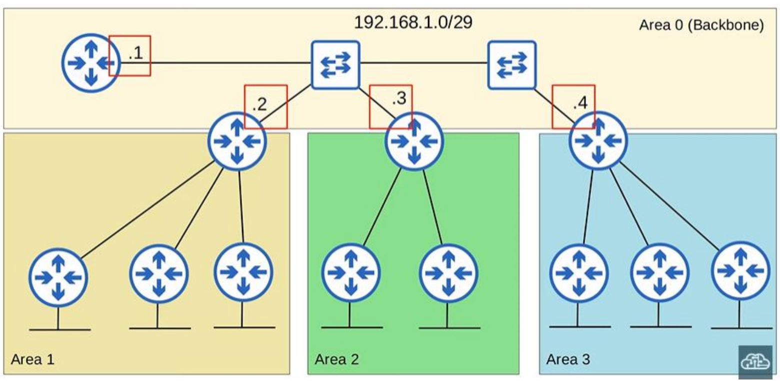


Even though all interfaces are in the same subnet and OSPF is enabled, the area 1 router will not become OSPF neighbours with the others.

Correct Neighbour Configuration:

This time, the area 1 ABR's interface in the 192.168.1.0/29 subnet is correctly configured in area 0.

Correct Example:



Now, all four routers will become OSPF neighbours.

Summary of OSPF Area Rules

Here's a summary of those three rules:

1. OSPF areas must be contiguous
2. Every area must have an ABR connected to area 0
3. Interfaces in the same subnet must be in the same area

Of course, many more OSPF concepts will be covered in the next few videos.

This will include:

- OSPF neighbours
- OSPF LSAs
- Additional OSPF details

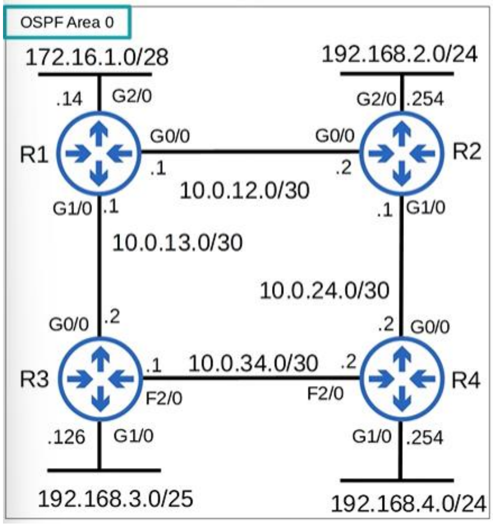
Basic OSPF Configuration (Single-Area OSPF)

Lab Topology Used

We will use the same network topology used previously for RIP and EIGRP.

All router interfaces are in OSPF area 0, because for the CCNA only single-area OSPF is required.

Network Topology:



R2, R3, and R4 are already configured.
We will configure OSPF on R1.

Entering OSPF Configuration Mode:

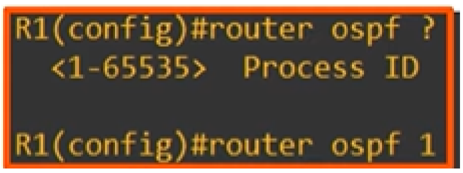
To enter OSPF configuration mode, use:

router ospf <process-id>

A router can run multiple OSPF processes at once.
The process ID is used only locally to identify them.

Typically, only one OSPF process is used.
Here, process ID 1 is used.

CLI:



```
R1(config)#router ospf ?
  <1-65535>  Process ID

R1(config)#router ospf 1
```

OSPF Process ID vs EIGRP AS Number:

In EIGRP, routers must have the same AS number to become neighbours.

In OSPF:

- The process ID is locally significant
- Routers with different process IDs can still become neighbours

For example:

- R1 could use process ID 1
- R2 could use process ID 2
- They would still become OSPF neighbours and exchange LSAs

The process ID is unrelated to the OSPF area.

Using the OSPF NETWORK Command:

OSPF uses wildcard masks, just like EIGRP.

Wildcard masks are inverse subnet masks.

When first attempting the NETWORK command:

```
R1(config-router)#network 10.0.12.0 0.0.0.3
% Incomplete command.
```

The command is incomplete because OSPF requires an area to be specified.

Activating OSPF on Interfaces:

OSPF is activated on interfaces by specifying the area.

All interfaces are configured in area 0.

CLI:

```
R1(config-router)#network 10.0.12.0 0.0.0.3 area 0
R1(config-router)#network 10.0.13.0 0.0.0.3 area 0
R1(config-router)#network 172.16.1.0 0.0.0.15 area 0
```

CCNA:

- Only **single-area OSPF** is required
- Area **0** is best practice
- Any area number would technically work

Function of the NETWORK Command:

The NETWORK command:

- Tells OSPF to look for interfaces with IP addresses in the specified range
- Activates OSPF on those interfaces in the specified area

It does not mean "advertise these networks".

Example:

- Network command: 10.0.12.0 0.0.0.3 area 0
- R1 G0/0 IP: 10.0.12.1
- Result: OSPF is activated on G0/0 in area 0

When OSPF is activated on an interface:

- The router tries to become OSPF neighbours with other OSPF routers
- R1 becomes neighbours with R2 and R3

PASSIVE-INTERFACE Command:

The PASSIVE-INTERFACE command works the same as in RIP and EIGRP.

It tells the router to:

- Stop sending OSPF hello messages out of an interface

OSPF uses hello messages to discover neighbours.

However:

- The router continues sending LSAs
- Neighbours are still informed about the subnet

CLI:

```
R1(config-router)#passive-interface g2/0
```

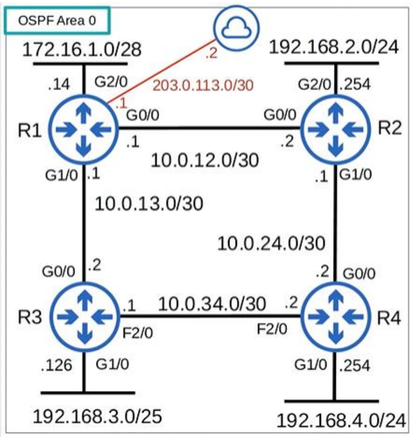
R1:

- Will not send hellos out of G2/0
- Will still advertise 172.16.1.0/28

This command should always be used on interfaces with no OSPF neighbours.

Advertising a Default Route into OSPF:

An Internet connection is added to R1.



A default route is configured with the ISP next-hop 203.0.113.2.

CLI (Default Route)

```
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
```

R1 Routing Table (Excerpt)

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

S* 0.0.0.0/0 via 203.0.113.2

```
Gateway of last resort is 203.0.113.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 203.0.113.2
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 10.0.12.2, 01:22:21, GigabitEthernet0/0
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/2] via 10.0.13.2, 01:22:11, GigabitEthernet1/0
      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/3] via 10.0.13.2, 00:00:06, GigabitEthernet1/0
                 [110/3] via 10.0.12.2, 00:00:06, GigabitEthernet0/0
      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
```

DEFAULT-INFORMATION ORIGINATE:

To advertise the default route into OSPF:

```
R1(config-router)#default-information originate
```

This causes:

- Creation of a new LSA
- Flooding of that LSA into OSPF

R2 Routing Table (Excerpt)

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 10.0.12.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 10.0.12.1, 00:01:38, GigabitEthernet0/0
```

O*E2 0.0.0.0/0 via 10.0.12.1

R3 and R4 also learn the default route.

SHOW IP PROTOCOLS (OSPF View)

CLI Output:

```
R1#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 172.16.1.14
  It is an autonomous system boundary router
 Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.12.0 0.0.0.3 area 0
    10.0.13.0 0.0.0.3 area 0
    172.16.1.0 0.0.0.15 area 0
  Passive Interface(s):
    GigabitEthernet2/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    4.4.4.4              110      00:00:08
    2.2.2.2              110      00:01:07
    3.3.3.3              110      00:01:07
    192.168.4.254        110      00:02:29
  Distance: (default is 110)
```

In Detail:

Routing Protocol is "ospf 1"

```
Routing Protocol is "ospf 1"
```

Routing for Networks:

10.0.12.0 0.0.0.3 area 0

10.0.13.0 0.0.0.3 area 0

172.16.1.0 0.0.0.15 area 0

Passive Interface(s):

GigabitEthernet2/0

The process ID is **1**, as configured earlier.

OSPF Router ID Selection:

Router ID selection priority:

1. Manually configured router ID
2. Highest IP on a loopback interface
3. Highest IP on a physical interface

Currently:

- No manual router ID
- No loopback interface
- Highest physical IP: 172.16.1.14

So R1’s router ID is 172.16.1.14.

```
Router ID 172.16.1.14
```

Manually Configuring Router ID:

```
R1(config-router)#router-id ?
  A.B.C.D  OSPF router-id in IP address format

R1(config-router)#router-id 1.1.1.1
% OSPF: Reload or use "clear ip ospf process" command, for this to take effect
```

The router displays:

Reload or use 'clear ip ospf process' command, for this to take effect

The router ID does not change immediately.

Clearing the OSPF Process:

To apply the new router ID:

```
R1#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
```

This resets OSPF on the router.

- This is a bad idea in production
- Routes are temporarily lost
- Acceptable in a lab

Afterward:

R1#show ip protocols

Router ID 1.1.1.1

```
R1#sh ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  It is an autonomous system boundary router
 Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.12.0 0.0.0.3 area 0
    10.0.13.0 0.0.0.3 area 0
    172.16.1.0 0.0.0.15 area 0
  Passive Interface(s):
    GigabitEthernet2/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2              110      00:01:40
    3.3.3.3              110      00:01:40
    4.4.4.4              110      00:01:40
  Distance: (default is 110)
```

Autonomous System Boundary Router (ASBR)   :

It is an autonomous system boundary router

```
It is an autonomous system boundary router
```

An ASBR:

- Connects the OSPF network to an external network

R1 becomes an ASBR because:

- It connects to the Internet
- DEFAULT-INFORMATION ORIGINATE is used

Number of Areas:

Output shows:

Number of areas in this router is 1

1 normal 0 stub 0 nssa

```
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```

This router is in only one area, because this is single-area OSPF.

Maximum Paths:

Default: Maximum paths is 4

OSPF:

- Does not support unequal-cost load balancing
- Supports ECMP up to 4 paths by default

To change:

CLI:

```
R1(config-router)#maximum-paths ?
  <1-32>   Number of paths

R1(config-router)#maximum-paths 8
```

Routing for Networks Section:

This section lists:

- The NETWORK commands used

These commands:

- Only determine which interfaces run OSPF
- Do not control LSA advertisement

```
Routing for Networks:
  10.0.12.0 0.0.0.3 area 0
  10.0.13.0 0.0.0.3 area 0
  172.16.1.0 0.0.0.15 area 0
```

OSPF Neighbours:

SHOW IP PROTOCOLS displays neighbours.

Router IDs shown:

- R2, R3, R4 use loopback IPs
- Loopback IPs became router IDs

```
Routing Information Sources:
  Gateway         Distance      Last Update
  2.2.2.2              110      00:01:40
  3.3.3.3              110      00:01:40
  4.4.4.4              110      00:01:40
```

OSPF Administrative Distance:

Default OSPF AD: Distance: (default is 110)

To change it:

```
R1(config-router)#distance ?
  <1-255>   Administrative distance

R1(config-router)#distance 85
```

Now:

- OSPF routes are preferred over EIGRP routes on this router